


**Figure 8.3 Cities and warehouses from figure 8.1, now *projected* on a 1-D space, a line segment.**

Figure 8.3 might help you picture the scenario: we have a line segment, with a starting point marked as `0`, and the cities and warehouses that we have seen in figure 8.1, but now all of them are lying on the same line. This is an approximation of the initial scenario, where those points belong the a 2-D plane, which in turn is an approximation of the reality, where the same points are laying on a 3-D curve surface. Depending on the use case, we might be fine with either approximation, or require a more precise model taking into account the curvature of Earth's surface.

Given a random point on the segment, we would like to know to which one of the reference points it is closer.

Being in a 1-D case, this looks awfully similar to binary search, right? Check figure 8.4 to see how binary search can be used to find the closest 1-D point.